

CULLMAN 3 ENE, AL, USA

WMO: 720010

Lat: 34.1954N

Lon: 86.7963W

Elev: 800

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.6	(c) 21.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 17.1	(k) 46.5	(l) 15.2	(m) 46.2	(n) 5.0	(o) 10	(p) 0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.0	(c) 92.9	(d) N/A	(e) 90.8	(f) N/A	(g) 88.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.7	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.0	(b) 12.2	(c) 10.6	(e) 11.0	(f) 95.5	(g) 4.6	(h) 3.2	(i) 7.7	(j) 97.8	(k) 5.0	(l) 99.6	(m) 2.4	(n) 101.4	(o) -1.0	(p) 103.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	40.3	44.9	52.4	59.8	67.8	75.6	77.6	77.5	72.2	60.9	49.8	45.3		
	DBStd	15.18	10.80	10.42	9.35	7.43	6.32	3.59	3.33	3.69	5.63	8.26	8.38	9.61		
	HDD50	970	338	206	87	12	1	0	0	0	0	11	103	212		
	HDD65	3227	766	566	397	187	49	0	0	0	11	181	458	612		
	CDD50	4769	37	63	161	305	551	768	856	851	666	347	98	66		
	CDD65	1551	0	2	6	29	135	318	391	386	227	52	4	1		
	CDH74	13860	0	6	67	292	1235	2997	3478	3388	1895	483	18	1		
	CDH80	5277	0	0	5	24	328	1229	1436	1453	691	109	2	0		
(14) Wind	WSAvg	4.6	5.5	5.7	5.7	5.3	4.2	3.9	3.7	3.5	3.6	4.2	4.7	5.2		
(15) Precipitation	PrecAvg	61.3	5.8	5.6	6.5	5.2	5.4	4.6	4.9	3.7	5.1	3.6	4.9	5.9		
	PrecMax	82.0	10.8	13.9	18.1	13.3	12.2	12.5	10.3	8.7	14.2	10.2	12.9	14.3		
	PrecMin	39.8	1.3	0.5	1.4	0.4	1.7	0.6	0.9	0.4	0.1	0.1	1.7	1.2		
	PrecStd	9.6	2.5	2.8	3.1	2.7	2.5	2.6	2.4	2.0	2.9	2.4	2.1	3.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.2	74.4	80.4	83.0	88.9	94.0	95.2	97.4	92.7	86.8	76.3	71.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	65.0	69.3	75.4	80.0	86.1	91.3	91.9	93.1	89.8	82.8	71.5	68.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	61.7	66.3	71.9	77.7	83.8	89.3	89.7	90.3	87.2	79.5	68.4	64.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	58.2	62.5	68.4	74.7	81.3	87.1	87.5	87.8	84.3	75.8	65.4	61.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	20.0	21.5	23.3	21.6	21.4	19.0	19.4	20.8	22.9	22.7	18.5		
		MCDBR	21.4	23.2	25.1	25.8	25.1	24.8	22.3	23.4	24.1	26.6	25.6	20.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.321	0.361	0.394	0.445	0.475	0.518	0.520	0.426	0.373	0.337	0.317		
	taud		2.510	2.492	2.398	2.295	2.196	2.147	2.048	2.030	2.283	2.416	2.459	2.521		
	Ebn at Noon		285	295	289	283	269	259	247	245	265	272	273	277		
	Edh at Noon		27	30	36	42	46	49	53	54	40	32	28	25		
(44) All-Sky Solar Radiation	RadAvg		799	964	1332	1721	1882	1970	1889	1751	1524	1244	911	683		
	RadStd		62	123	124	121	147	155	112	96	135	160	98	66		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+259	+189

Nomenclature: See separate page